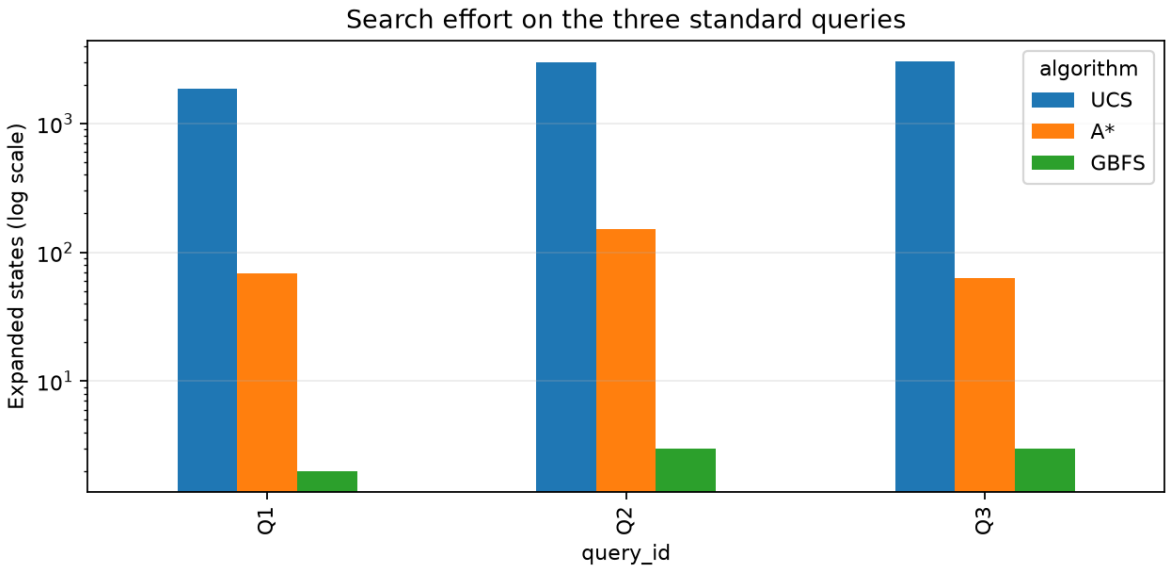
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RESULT

This section presents the results obtained from the implementation of UCS, GBFS, A*, and Weighted A* algorithms. The evaluation focuses on route cost, search efficiency, and heuristic admissibility using three airport route queries: KUL–KEF, SIN–EZE, and NRT–CPT.



HEURISTICS

query_id	airport	goal	haversine_km	shortest_route_km	slack_km	admissible
Q1	GLA	KEF	1347.458656	1347.458656	0	TRUE
Q1	IBZ	KEF	3206.354488	3244.496017	38.1415289	TRUE
Q1	GZP	KEF	4744.908783	4745.73721	0.828426737	TRUE
Q1	TUU	KEF	5710.898378	5847.10014	136.2017626	TRUE
Q1	OAK	KEF	6731.273796	6893.044504	161.770708	TRUE
Q1	ANC	KEF	5424.449731	8136.991616	2712.541885	TRUE
Q1	JSR	KEF	8696.844768	9238.71883	541.8740616	TRUE
Q1	AKY	KEF	9162.994187	10503.69942	1340.705232	TRUE
Q1	ZCO	KEF	12238.32244	12935.7591	697.4366678	TRUE
Q1	OTD	KEF	7561.169048	30818.5261	23257.35705	TRUE
Q2	MVD	EZE	228.6429836	228.6429836	0	TRUE
Q2	GUA	EZE	6448.41046	6462.039593	13.62913259	TRUE
Q2	SBN	EZE	8969.088441	8973.011367	3.92292531	TRUE
Q2	DIK	EZE	10095.49327	10307.23177	211.7385024	TRUE
Q2	LAU	EZE	10724.0417	11502.11859	778.076889	TRUE
Q2	BMA	EZE	12583.76859	12584.10015	0.331556801	TRUE
Q2	KUT	EZE	13341.02797	13831.40497	490.3770009	TRUE
Q2	NCU	EZE	14697.0819	15633.03718	935.9552816	TRUE
Q2	MIG	EZE	18414.83067	18788.05667	373.2259947	TRUE
Q2	DEE	EZE	11609.54289	29101.91686	17492.37397	TRUE
Q3	GRJ	CPT	348.2713383	348.2713383	0	TRUE
Q3	EBL	CPT	8237.783124	8657.320648	419.5375244	TRUE
Q3	SCN	CPT	9318.788423	9648.053521	329.2650977	TRUE
Q3	BGO	CPT	10552.3895	10552.81996	0.430464345	TRUE
Q3	ATM	CPT	8044.526554	11640.96612	3596.439564	TRUE
Q3	SHA	CPT	12951.23342	13083.16875	131.9353303	TRUE
Q3	ULG	CPT	11599.70321	14080.55073	2480.847513	TRUE
Q3	YNG	CPT	13117.55468	14938.90845	1821.353774	TRUE
Q3	APW	CPT	14583.12755	16674.4641	2091.336554	TRUE
Q3	OTD	CPT	11243.53624	37413.65774	26170.1215	TRUE

METRICS

query_id	algorithm	hops	cost_km	expanded	runtime_ms	path	gap_pct
Q1	UCS	4	11461.34858	1851	13.6897	KUL -> HKT -> LED -> HEL -> KEF	0
Q1	GBFS	2	12500.49849	2	0.5712	KUL -> LHR -> KEF	9.066558832
Q1	A*	4	11461.34858	69	1.0205	KUL -> HKT -> LED -> HEL -> KEF	0
Q2	UCS	3	17817.35625	3002	21.4396	SIN -> JNB -> GRU -> EZE	0
Q2	GBFS	3	17817.35625	3	0.8695	SIN -> JNB -> GRU -> EZE	0
Q2	A*	3	17817.35625	151	1.2822	SIN -> JNB -> GRU -> EZE	0
Q3	UCS	3	14813.31926	3042	13.6577	NRT -> HKG -> MRU -> CPT	0
Q3	GBFS	3	15811.29464	3	0.3767	NRT -> DOH -> JNB -> CPT	6.737013856
Q3	A*	3	14813.31926	63	1.422	NRT -> HKG -> MRU -> CPT	0
Q1	Weighted A* (1.5h)	2	12274.31753	2	0.9631	KUL -> AMS -> KEF	7.093135224
Q2	Weighted A* (1.5h)	3	17817.35625	3	0.499	SIN -> JNB -> GRU -> EZE	0
Q3	Weighted A* (1.5h)	4	14929.35397	4	0.3147	NRT -> CMB -> SEZ -> JNB -> CPT	0.783313325

This experiment successfully implemented and evaluated four informed and uninformed search approaches for finding routes within an airline network: Uniform Cost Search, Greedy Best-First Search, A* Search, and Weighted A* Search. The results demonstrate that UCS consistently identified the minimum-cost routes but required the largest number of expanded states. A* achieved the same route costs as UCS across all three tested queries while reducing the average number of expanded states by 96.4%. This indicates that A* provided a more efficient search process without sacrificing route optimality in the tested cases.

GBFS explored the fewest states but did not consistently identify the shortest routes because it considered only the heuristic estimate. Meanwhile, Weighted A* demonstrated that increasing the heuristic weight can reduce search effort but may result in higher route costs. The Haversine heuristic recorded zero admissibility violations across 30 sampled cases. Although this supports its suitability for the tested network, the result does not establish global admissibility mathematically.

Overall, the experiment achieved its objectives by demonstrating the differences between the search algorithms in terms of route cost, search efficiency, and heuristic performance. Among the evaluated methods, A* provided the most balanced performance by maintaining optimal route costs while substantially reducing the number of expanded states compared with UCS.